

```

% --- MISSION 1: Baseline Re-entry Flow ---
% This script visualizes the uniform plasma flow during atmospheric
% re-entry before the thermal protection shield is activated.

clear; clc; close all;

%% Step 1: Define the Simulation Space
% 'meshgrid' creates a 2D grid of X and Y coordinates for our plot.
[X, Y] = meshgrid(-2:0.2:2); % Creates a grid from -2 to 2

%% Step 2: Define the Plasma's Velocity
% For re-entry, the plasma moves towards the vehicle. We'll model this
% as a uniform flow moving downwards in the y-direction.
u = zeros(size(X)); % 'u' is the horizontal (x) velocity, which is zero.
v = -ones(size(Y)); % 'v' is the vertical (y) velocity, which is -1 (downwards).

%% Step 3: Visualize the Flow Field
% The 'quiver' function draws an arrow at each grid point to show the flow.
figure('Name', 'MHD Mission 1: Baseline', 'Position', [100, 100, 800, 600]);
quiver(X, Y, u, v, 'b'); % Plot the vectors as blue arrows

%% Step 4: Add Professional Formatting
hold on; % Allows us to add more to the plot

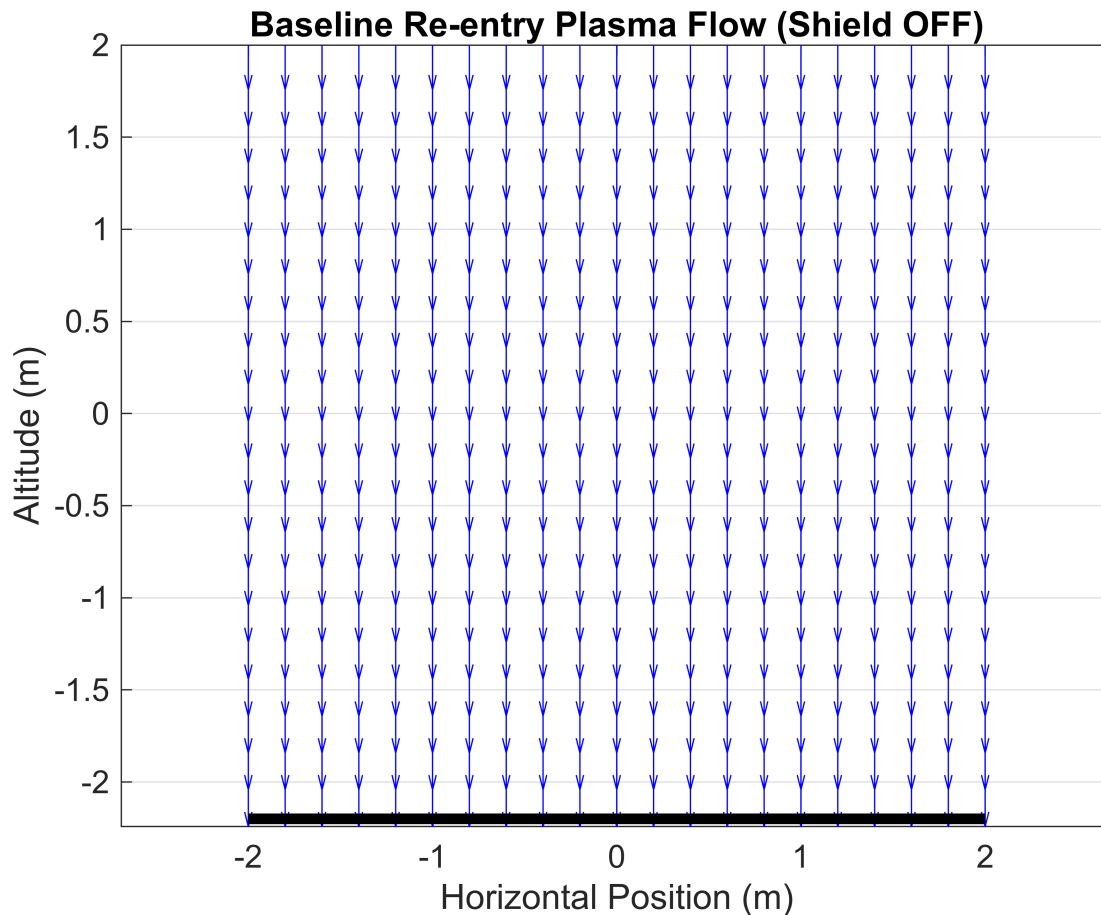
% Add a thick black line to represent the spacecraft surface
plot([-2, 2], [-2.2, -2.2], 'k', 'LineWidth', 4);

hold off; % Done adding to the plot

% Add titles and labels for clarity
title('Baseline Re-entry Plasma Flow (Shield OFF)');
xlabel('Horizontal Position (m)');
ylabel('Altitude (m)');
axis equal; % Ensures the arrows are not distorted
grid on;
set(gca, 'FontSize', 12);

%% Step 5: Export a High-Quality Figure
exportgraphics(gcf, 'MHD_Mission1_Baseline.pdf', 'Resolution', 300);

```



```
% --- MISSION 2: Defining the Magnetic Field ---
% This script models and visualizes the strength of the magnetic field
% that will be used to deflect the plasma.

clear; clc; close all;

%% Step 1: Define the Simulation Space (Same as Mission 1)
[X, Y] = meshgrid(-2:0.2:2);

%% Step 2: Define the Magnetic Field Source
% We will model a source located below the spacecraft surface.
source_pos = [0, -2.5]; % Position of the B-field source [x, y]
B_strength = 50;        % An arbitrary strength for the field at its source

%% Step 3: Calculate the Field Strength at Every Point
% The magnetic field (Bz, pointing out of the screen) gets weaker with
% the square of the distance from the source (the inverse-square law).
dist_sq = (X - source_pos(1)).^2 + (Y - source_pos(2)).^2;
Bz = B_strength ./ dist_sq;

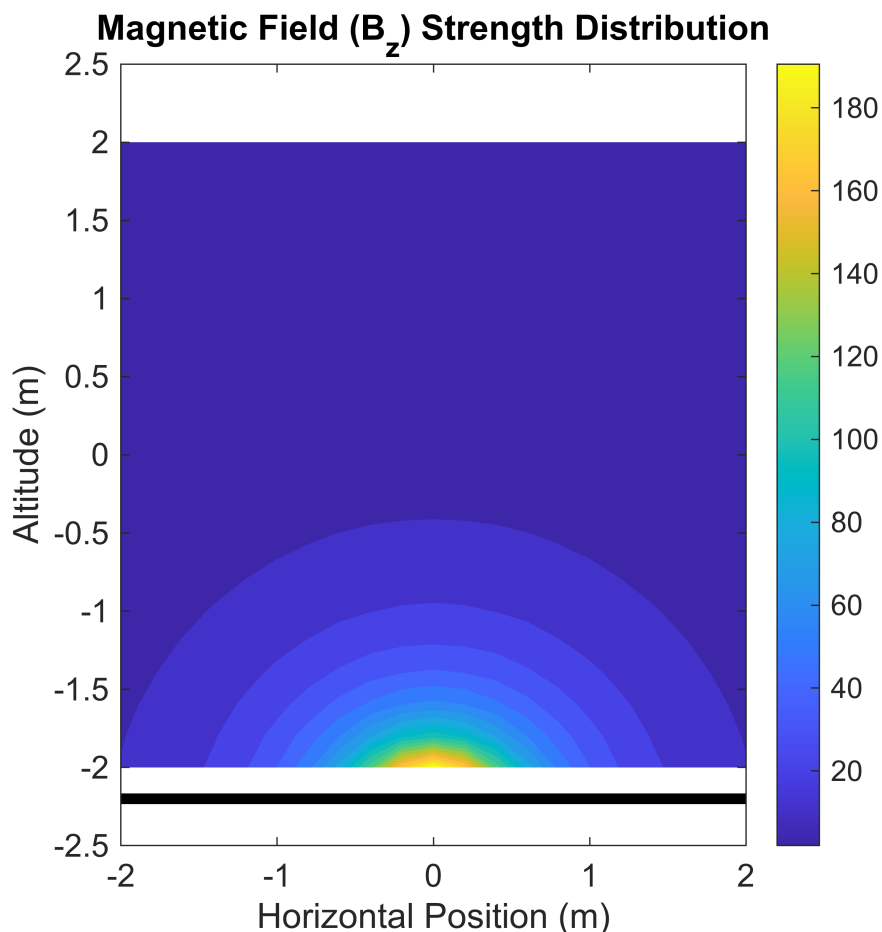
%% Step 4: Visualize the Magnetic Field Strength
% A filled contour plot is excellent for showing field strength.
```

```
figure('Name', 'MHD Mission 2: B-Field', 'Position', [100, 100, 800, 600]);
contourf(X, Y, Bz, 20, 'LineStyle', 'none'); % 20 contour levels, no lines
colorbar; % Adds a color bar to show the scale of the field strength

hold on;
% Add the spacecraft surface for context
plot([-2, 2], [-2.2, -2.2], 'k-', 'LineWidth', 4);
hold off;

title('Magnetic Field (Bz) Strength Distribution');
xlabel('Horizontal Position (m)');
ylabel('Altitude (m)');
axis equal; axis([-2 2 -2.5 2.5]);
set(gca, 'FontSize', 12);

%% Step 5: Export a High-Quality Figure
exportgraphics(gcf, 'MHD_Mission2_B-Field.pdf', 'Resolution', 300);
```



```
% --- MISSION 3: Calculating the Deflection ---
% This script calculates the Lorentz force on the plasma at every point
% and determines the new, deflected velocity field. The results are saved
% to a file for the final mission.
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```
clear; clc; close all;
```

```
%% Step 1: Define the Baseline Flow and Magnetic Field
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% This section combines the setup from Mission 1 and Mission 2.
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[X, Y] = meshgrid(-2:0.2:2);
```

```
u_original = zeros(size(X));
```

```
v_original = -ones(size(Y));
```

```
source_pos = [0, -2.5];
```

```
B_strength = 50;
```

```
dist_sq = (X - source_pos(1)).^2 + (Y - source_pos(2)).^2
```

```
dist_sq = 21x21
```

4.2500	3.4900	2.8100	2.2100	1.6900	1.2500	0.8900	0.6100 ...
4.4900	3.7300	3.0500	2.4500	1.9300	1.4900	1.1300	0.8500
4.8100	4.0500	3.3700	2.7700	2.2500	1.8100	1.4500	1.1700
5.2100	4.4500	3.7700	3.1700	2.6500	2.2100	1.8500	1.5700
5.6900	4.9300	4.2500	3.6500	3.1300	2.6900	2.3300	2.0500
6.2500	5.4900	4.8100	4.2100	3.6900	3.2500	2.8900	2.6100
6.8900	6.1300	5.4500	4.8500	4.3300	3.8900	3.5300	3.2500
7.6100	6.8500	6.1700	5.5700	5.0500	4.6100	4.2500	3.9700
8.4100	7.6500	6.9700	6.3700	5.8500	5.4100	5.0500	4.7700
9.2900	8.5300	7.8500	7.2500	6.7300	6.2900	5.9300	5.6500
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```
Bz = B_strength ./ dist_sq;
```

```
%% Step 2: Calculate the Lorentz Force ( $F = q * v \times B$ )
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```
% Assuming charge  $q = 1$  for simplicity.
```

```
%  $v = [u, v, 0]$  and  $B = [0, 0, B_z]$ 
```

```
% The cross product simplifies to:
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```
%  $F_x = q * (v_y * B_z)$ 
```

```
%  $F_y = q * (-v_x * B_z)$ 
```

```
Fx = v_original .* Bz;
```

```
Fy = -u_original .* Bz; % This will be zero since u_original is zero
```

```
%% Step 3: Calculate the New, Deflected Velocity Field
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```
% We add a fraction of the force to the original velocity to simulate
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% how the force pushes the plasma particles onto a new path.
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```
deflection_factor = 0.05; % Controls how strong the deflection effect is
```

```
u_deflected = u_original + Fx * deflection_factor
```

```
u_deflected = 21x21
```

-0.5882	-0.7163	-0.8897	-1.1312	-1.4793	-2.0000	-2.8090	-4.0984 ...
-0.5568	-0.6702	-0.8197	-1.0204	-1.2953	-1.6779	-2.2124	-2.9412
-0.5198	-0.6173	-0.7418	-0.9025	-1.1111	-1.3812	-1.7241	-2.1368
-0.4798	-0.5618	-0.6631	-0.7886	-0.9434	-1.1312	-1.3514	-1.5924
-0.4394	-0.5071	-0.5882	-0.6849	-0.7987	-0.9294	-1.0730	-1.2195
-0.4000	-0.4554	-0.5198	-0.5938	-0.6775	-0.7692	-0.8651	-0.9579
-0.3628	-0.4078	-0.4587	-0.5155	-0.5774	-0.6427	-0.7082	-0.7692
-0.3285	-0.3650	-0.4052	-0.4488	-0.4950	-0.5423	-0.5882	-0.6297
-0.2973	-0.3268	-0.3587	-0.3925	-0.4274	-0.4621	-0.4950	-0.5241
-0.2691	-0.2931	-0.3185	-0.3448	-0.3715	-0.3975	-0.4216	-0.4425

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```
v_deflected = v_original + Fy * deflection_factor
```

```
v_deflected = 21x21
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1 ...
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
-1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1    -1
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```

```
%% Step 4: Save All Results for the Final Mission
% The 'save' command creates a file containing all the specified variables.
save('mhd_data.mat', 'X', 'Y', 'u_original', 'v_original', 'u_deflected',
'v_deflected');

% Display a confirmation message in the Command Window
fprintf('Mission 3 Complete.\n');
```

Mission 3 Complete.

```
fprintf('All necessary data has been calculated and saved to "mhd_data.mat".\n');
```

All necessary data has been calculated and saved to "mhd_data.mat".

```
fprintf('You are now ready for the final mission.\n');
```

You are now ready for the final mission.

```
% --- MISSION 4: Final MHD Shield Visualization ---
% This script loads the pre-calculated data and generates the final
% side-by-side comparison plot of the MHD shield's effect.
```

```
clear; clc; close all;
```

```
%% Step 1: Load the Data from Mission 3
% The 'load' command imports all variables from the specified .mat file
% into the current workspace.
load('mhd_data.mat');
```

```
%% Step 2: Create the Figure and Main Title
figure('Name', 'MHD Shield Simulation: Final Result', 'Position', [100, 100, 800,
700]);
% 'sgtitle' creates a centered title for the entire figure.
sgtitle('MHD Thermal Protection System (TPS) Simulation', 'FontSize', 16,
'FontWeight', 'bold');
```

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%% Step 3: Create the Top Plot (Shield OFF)
% 'subplot' divides the figure window into a grid. (2, 1, 1) means a
% 2-row, 1-column grid, and we are now working on the 1st plot.
subplot(2, 1, 1);
quiver(X, Y, u_original, v_original, 'b'); % Plot the original baseline flow
hold on;
plot([-2, 2], [-2.2, -2.2], 'k', 'LineWidth', 4); % The spacecraft surface
hold off;
title('Baseline Re-entry Plasma Flow (Shield OFF)');
ylabel('Altitude (m)');
axis equal; axis([-2 2 -2.5 2.5]);
grid on;
set(gca, 'FontSize', 12);

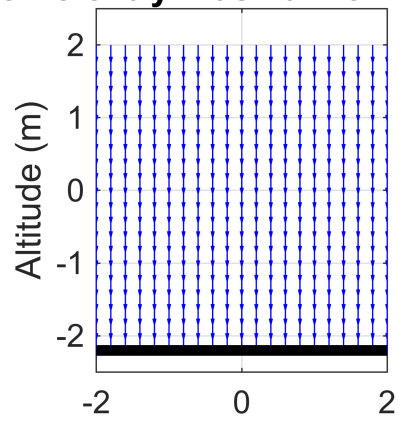
%% Step 4: Create the Bottom Plot (Shield ON)
% Now we activate the 2nd plot in our grid.
subplot(2, 1, 2);
quiver(X, Y, u_deflected, v_deflected, 'r'); % Plot the NEW deflected flow
hold on;
plot([-2, 2], [-2.2, -2.2], 'k-', 'LineWidth', 4); % The spacecraft surface
hold off;
title('Deflected Plasma Flow (MHD Shield ACTIVE)');
xlabel('Horizontal Position (m)');
ylabel('Altitude (m)');
axis equal; axis([-2 2 -2.5 2.5]);
grid on;
set(gca, 'FontSize', 12);

%% Step 5: Export the Final, High-Quality Figure
exportgraphics(gcf, 'MHD_Mission_Final_Result.pdf', 'Resolution', 300);

```

MHD Thermal Protection System (TPS) Simulation

Baseline Re-entry Plasma Flow (Shield OFF)



Deflected Plasma Flow (MHD Shield ACTIVE)

